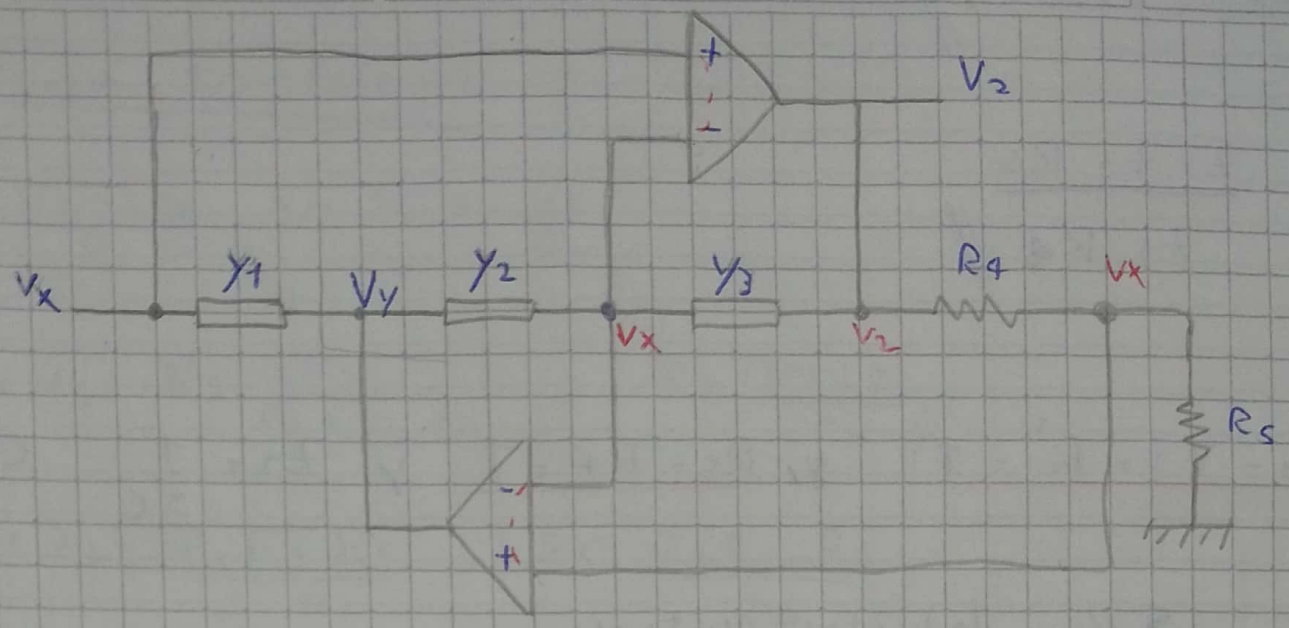

① $(V_y - V_x)\,Y_2 = (V_x - V_2)\,Y_3$

② $V_x = V_2 \dfrac{R_s}{R_4 + R_s} \longrightarrow V_2 = V_x \dfrac{R_4 + R_s}{R_s}$

③ $I_x = (V_x - V_y)\,Y_1$

① $V_y Y_2 - V_x Y_2 = V_x Y_3 - V_2 Y_3$

$$V_y Y_2 = V_x Y_3 - V_x \frac{R_4 + R_s}{R_s} Y_3 + V_x Y_2$$

$$Y_2 V_y = V_x \left(Y_3 - \frac{R_4 + R_s}{R_s} Y_3 + Y_2 \right)$$

$$V_y = V_x \left(\frac{\cancel{Y_3 R_s} - R_4 Y_3 - \cancel{R_s Y_3} + Y_2 R_s}{R_s Y_2} \right)$$

③ $$I_x = V_x Y_1 - V_x \left(\frac{Y_2 R_s - R_4 Y_3}{R_s Y_2} \right) Y_1$$

$$I_x = V_x \left(Y_1 - \frac{Y_2 R_s Y_1 - R_4 Y_3 Y_1}{R_s Y_2} \right)$$

$$I_x = V_x \left(\frac{\cancel{Y_1 Y_2 R_s} - \cancel{Y_1 Y_2 R_s} + Y_1 Y_3 R_4}{R_s Y_2} \right)$$

Papel de fibra de caña de azúcar.

Escaneado con CamScanner

$$Z_x = \frac{V_x}{I_x} = \frac{Y_2 R_5}{Y_1 Y_3 R_4} //$$

$$Z_x = \frac{Z_1 Z_3 R_5}{Z_2 R_4} //$$

Si $Z_1 = Z_3 = R' = 1$ y $R'_5 = R'_4 = 1$ y $Z_2 = \frac{1}{sC}$, $C' = 1$

$$Z_x = s //$$ Inductor unitario

Escaneado con CamScanner